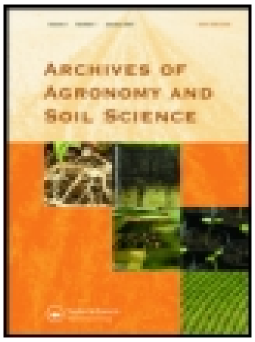

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Variability of soil water content controlled by evapotranspiration and groundwater–root zone interaction

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ABSTRACT

The central moments of soil water content (SWC) variability at the field scale are determined by soil texture, considering both smooth topography and groundwater table position. The characteristics of variability are governed by other soil factors like soil structure, micro relief, preferred water flow paths, root system characteristics, rock content, etc. This paper shows the integral effect of all these hardly quantifiable factors on SWC variability simulated by the processes of evapotranspiration and groundwater–root zone interaction using the HYDRUS ET model. SWC and soil hydraulic characteristics were spatially determined over a 4.5 ha field during two sampling campaigns under different atmospheric and groundwater conditions, and data distributions were compared to SWC distributions provided by mathematical modeling. The entire spring–summer period of 2003 was then examined for changes of SWC spatial variability. It was found that evapotranspiration influences SWC spatial variability only if SWC is under the critical value when wetter parts of the field evaporate more water than drier parts, resulting in smoothed SWC variability. Under wet conditions the spatial variability of SWC increases by drainage, as those parts of the soil with coarser texture drain faster than finer-textured parts.

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Evapotranspiration; drainage; soil water content; spatial variability

Introduction

Soil water content (SWC) is an integral compound of the hydrological balance at certain locations, including all relevant factors and processes active at the wide range of spatial and temporal scales. Thus, it is highly variable spatially (Western & Grayson 1998). Soil moisture is an important parameter regarding land use and agriculture – it determines optimal soil conditions for soil tillage and influences microorganisms and enzyme activity. Spatial patterns of soil moisture determine solute transport, erosion processes (Grayson et al. 1997), flooding, and land–atmosphere interactions (Georgakakos & Baumer 1996 in Western & Blöschl 1999). Improved understanding of the spatial distribution of soil water content (SWC) is therefore required for a range of applications in water-related disciplines. There is evident discrepancy in the literature related to the relationship between mean value and variance of soil moisture fields. Some studies found an increase in SWC variance with decreasing mean SWC value, while others showed the opposite trend. It is time-consuming and labor intensive to measure SWC with a high sampling frequency and over a large spatial extent. Therefore, understanding the principles of SWC spatial variability may help us to save on time and effort connected with its detection. The observations of authors working on the spatial variability of SWC at different scales can be summarized as follows:

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(1) SWC data usually show normal spatial distribution if the stationarity of determinant factors at a particular scale is fulfilled (Hills & Reynolds 1969; Nielsen et al. 1973; Warrick et al. 1977). Loague (1992) observed the normal distribution of SWC for two data sets (each containing 50 samples) collected along two transects. However, he did not confirm the normal distribution for the 157 samples data set collected from the regular grid covering the whole (0.1 km^2) research catchment area. Western et al. (1998) observed seasonal changes in the shape of soil moisture distribution function. They found normal distribution only during dry periods when the soil moisture is typically random, not determined by topography. A normal distribution of topsoil water content was reported by Bell et al. (1980), Hawley et al. (1983), Francis et al. (1986) and Nyberg (1996).
(2) The coefficient of variation (CV) as a measure of the relative spatial variability of SWC in the upper 30 cm horizon has values ranging between 10–30 %. There is a tendency for lower average values of SWC being related to higher values of CV.
(3) Drainage generally enhances the variability of SWC, since relatively finer-textured soil (with usually greater porosity) drains more slowly than coarser-textured soil (with less porosity). The contribution of evapotranspiration to the variability of SWC, on the other hand, is not negligible. When the soil is wet there is no limit to evaporation, which and is close to the potential rate of control by actual atmospheric demand. Atmospheric demand for evaporation usually has small spatial variation at the field scale if landscape structures such as shelter belts or hedgerows are not present (Orfanus & Eitzinger 2010). Therefore, evaporation during this wet stage can only reduce mean SWC, but does not change its variance. In the next stage, SWC becomes lower than the threshold SWC and actual evapotranspiration becomes lower than potential. Evapotranspiration begins to be selective, i.e. wetter parts of the field evaporate more than drier parts, thus smoothing the variability of SWC.

At small scales where spatial variations of climate and vegetation can be ignored, the dominant factors controlling SWC distribution are soil properties and topography. When SWC data are collected over a relatively flat area, the topography can be assumed homogeneous and SWC variability has a quasi-stochastic character. The aim of this research was to examine how SWC variability at the scale of a 4.5 ha field is changed owing to dominant drying processes: soil water drainage, evapotranspiration and groundwater – vadose zone hydraulic interaction. We propose a method of using statistically processed hydrophysical characteristics of topsoil and subsoil as inputs to a 1-D mathematical model simulating the movement of water in an unsaturated soil profile.

Materials and methods

3 Stages of evapotranspiration induced drying of soil

Evaporation and drainage are two dominant factors controlling soil moisture dry-down. Studies frequently report that there are three stages of soil evaporation (e.g. Idso et al. 1974; Hillel 1998) as follows: stage I, the atmosphere-controlled; stage II, which is controlled by soil hydraulic properties; and stage III, which is atmospheric demand controlled (Pan & Peters-Lidard 2008). It is possible to explain the temporal changes in the mutual relationship between SWC variance and mean SWC by considering a subsequent dry-down process from the saturated SWC (θ_S) to the residual SWC (θ_r). Between those upper and lower limits of SWC status, two transition SWC states (denoted as θ_{la} and θ_{wp}) are used to define the three stages of the soil moisture dry-down process. Stage I is from θ_S to θ_{la} (SWC of limited availability), stage II from θ_{la} to θ_{WP} (wilting point) and stage III from θ_{WP} to θ_r.

These three stages can be illustrated in a plot of actual evapotranspiration (ET) rate vs. time (Hillel 1998) (Figure 1). In stage I, the ET rate is equal to the potential evapotranspiration rate (PET) and independent of SWC; in stage III, the ET rate is much lower than the potential rate (close to 0)

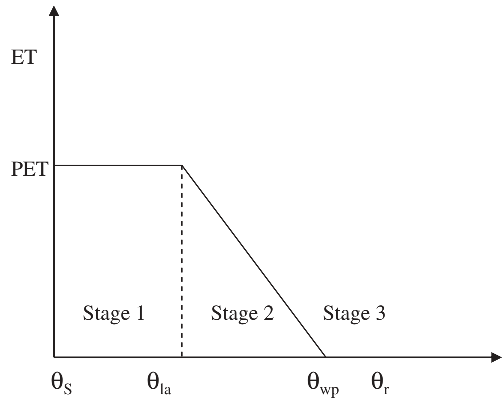

Figure 1. The actual evaporation rate as a function of soil moisture and the potential evaporation rate in the three stages of the soil dry-down process. θ_S is the saturated soil water content, θ_{la} is the soil water content of limited availability, θ_{wp} is the permanent wilting point and θ_r is the residual soil water content.

and approximately a constant; in stage II, the ET rate is proportional to SWC. θ_{la} and the slope of the proportionality in stage II are dependent on the PET on a particular day. To write these principles as equations, we express the actual evapotranspiration rate as follows:

$$\begin{gathered} \mathrm{ET}/\mathrm{ETP} = 1 \text{ for } \theta_{la} < \theta \\ \mathrm{ET}/\mathrm{ETP} = \alpha(\theta - \theta_{wp}) \text{ for } \theta_{wp} < \theta < \theta_{la} \\ \mathrm{ET}/\mathrm{ETP} = 0 \text{ for } \theta_r < \theta < \theta_{wp} \end{gathered} \tag{1}$$

Taking the left side of (Equation (1)) as equal to 1, it can be rewritten as:

$$\theta_{la} = 1/\alpha + \theta_{wp} \text{ for } \theta_{wp} < \theta < \theta_{la} \tag{2}$$

θ_{wp} is conventionally estimated from the soil water retention curve as θ (−1.5 MPa). θ_{la} and α are highly variable depending mostly on ETP and other characteristics like root system density and its spatial distribution. α for each sample is determined by:

$$\alpha = \alpha_m + (\alpha_1 - \alpha_m) \exp\left[-\delta(\mathrm{ETP} - 1)\right] \tag{3}$$

δ can be set to 0.25 for crops with dense root systems (Cowan 1965; Sudnicyn 1979), but α_m and α_1 have to be calculated.

To calculate α_m we used (Equation (2)), substituting θ_{la} with the saturated SWC as its theoretical maximum value. α_1 is the slope of Equation (1) if Ep = 1 mm day^{-1} (Novák 1986, 1989). Novák (1989) assumes (α_1–α_m) is constant for all soils and is equal to 12.3.

Equations (2) and (3) emphasize the fact that the critical SWC of limited availability (θ_{la}), below which the relative evapotranspiration drops below 1 is not a stable characteristics of a particular soil but changes over time with potential evapotranspiration and root system development.

The research plot and soil sampling

The research was conducted on a 4.5 ha (150 × 300 m) plot situated near the village of Moravský Svätý Ján (MSJ) in the Záhorská Lowland in southwestern Slovakia. The soil substrate in this area belongs to alluvium of the River Moravia. The plot has a uniform tillage, but marked textural differences in soil. The investigated field was ploughed for decades until 2001, after which minimum tillage was applied. Local soils are Arenic Regosol covering an undivided 60% of the plot area with sandy texture, and Mollic Gleysol extending on the remaining 40% (ISSS-ISRIC-FAO

1998). These soil taxons are divided by a narrow and easily identifiable borderline. The Arenic Regosol has a loamy sand texture and Mollic Gleysol consists of clay loam. A transition area of only a few meters around the borderline is texturally variable.

Two intensive sampling campaigns were performed in 2002–2003. The first sampling date (10 April 2002) was chosen to ensure bare soil surfaces and a low ETP rate to avoid significant water losses during the sampling period. The soil was sampled in a regular 20 × 20 m grid. Soil samples were taken from depths of 0.10–0.15 and 0.30–0.35 m and placed in stainless cylinders of 100 cm^3 volume and 5 cm height. Actual SWC was estimated gravimetrically as volume percentage, and the saturated hydraulic conductivities and drying branches of soil water retention curves, *θ(h)* were estimated in the laboratory. Pressure chambers (Santa Barbara, California Device) were used to estimate soil water content for characteristic values of soil water potential (hw = 0.06, 0.20, 0.8, 1.5, 3 and 15 bar). Saturated hydraulic conductivity was determined by the falling head method. Both actual and saturated SWC, as well as saturated hydraulic conductivity were determined on each soil sample (n = 128). Due to the limited capacity of the pressure chambers the soil water retention curves were measured only for 43 samples evenly selected from the whole research plot area and from both depths.

Mathematical simulations

We next determined the spatial variability of SWC reproduced by the variability of estimated *θ(h)* and *k(θ)* when using a 1-D mathematical model. The model HYDRUS–ET was used as a simulation tool (Šimůnek et al. 1997). This model uses a 1-D version of the Richards equation to solve the transport of water in the soil profile:

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial z}\left[D(\theta)\frac{\partial\theta}{\partial z}\right] - \frac{\partial k\partial\theta}{\partial\theta\partial z} \tag{4}$$

The basic hydraulic functions – soil water retention curve and hydraulic conductivity function – are expressed according to van Genuchten (1980):

$$\theta\,(h) = \theta_r + \frac{\theta_s - \theta_r}{\left[1 + |\alpha h|^n\right]^m}\ \text{for}\, h < 0$$
$$\theta\,(h) = \theta_s\ \text{for}\ h \geq 0 \tag{5}$$

and

$$K\,(h) = K_s\,S_e^l\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2 \tag{6}$$

where

$$m = 1 - 1/n\,, n{>}1$$

$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}$ -relative SWC ()

θ_s– volumetric SWC at saturation (cm^3 cm^{-3})

θ_r– residual volumetric SWC (cm^3 cm^{-3})

K_s– saturated hydraulic conductivity (cm day^{-1})

h– soil water potential (cm)

α, n– van Genuchten's parameters (cm^{-1}), ().

θ(h) and *K(h)* determined on samples from the 0.10–0.15 m horizon were assigned for the soil layer 0–0.25 m, and those estimated from horizon 0.30–0.35 m were assigned for the remainder of the soil profile during the simulation period.

The upper boundary condition was defined by daily meteorological characteristics (air temperature, precipitation, wind speed, sunshine duration and relative humidity) measured at the climatological station in MSJ (48°34′54′′ N, 16° 59′39′′E, 155 m above sea level). The weekly averages of groundwater table depth were set as bottom boundary condition. The first day of mathematical simulation was the first day of the year with solely positive temperatures (16 February 2002). The initial condition was set as soil water potential evenly distributed within the soil profile, with zero value of the pressure head at a depth of 0.70 m (groundwater level on February 16 2002). Measured and simulated SWC distributions to 10 April 2002 were compared to validate model performance.

The model gives a schematic of the stress response function as used by Feddes et al. (1978). Water uptake is assumed to be zero close to saturation (i.e. wetter than some arbitrary anerobiosis point, P_0). Root water uptake is also zero for pressure heads less than the wilting point (P_3). Water uptake is considered optimal between pressure heads P_{opt} and P_2, whereas for pressure heads between P_2 and P_3 (or P_0 and P_{opt}) water uptake decreases (or increases) linearly with pressure head. The P_2 value of the limiting pressure head below which roots cannot extract water at the maximum rate is specified for a maximum potential transpiration rate of 5 mm day^{-1} (P_{2H}) and minimum potential transpiration rate of 1 mm day^{-1} (P_{2L}). HYDRUS–ET currently implements the same linear interpolation scheme as used in several versions of the SWATRE code (e.g. Wesseling & Brandyk 1985).

Another option, the S-shaped function (van Genuchten 1985), is determined by the exponent, P_0 [-], in the root water uptake response function due to water stress (its recommended value is 3), and P_{50} the coefficient [L] given by the pressure head value when water uptake is reduced by 50%.

The Verhulst–Pearl logistic growth function is used to describe root growth during the growing season. To perform the simulation, the initial root growth time and harvest time, as well as the initial and maximum rooting depths must be specified. The root growth factor is calculated under the assumption that 50% of the rooting depth is reached at the midpoint of the growing season. The roots are assumed to be exponentially distributed with depth.

A total of 270 HYDRUS–ET model simulations were performed according to the algorithm, as follows.

(1) First, the performance of the HYDRUS–ET model was tested on four specific soil profiles in the MSJ experimental field. Two profiles were on the clay loam part of the field and the other two on loamy sand. The main hydrophysical characteristics (*θ(h)* and *K(h)*) were determined for the topsoil and subsoil of these four soil profiles. The simulation period was 16 February–10 April 2002.
(2) The 43 soil water retention functions (*(θ(h)*) were statistically processed in such a way that for each particular value of soil water potential (0.06, 0.20, 0.8, 1.5, 3 and 15 bar), the average SWC and its standard deviation were estimated in each textural class separately. Then for each textural class we calculated the average soil water retention curve and two contour curves (average + standard deviation (SD) and average SD) by adding or subtracting the standard deviation from the average value, as presented in Figure 2.
(3) These six sets of *θ(h)* parameters (θ_S, θ_r, a, n) were then combined for topsoil and subsoil and together with 32 values of the saturated hydraulic conductivity, K_S were chosen from the original 128 measured values keeping their frequency distributions within the particular textural classes. A total of 270 simulations were performed for the spring–summer period (16 February–2 July 2003). During this period, the root water uptake of maize (*Zea mays*) was also considered in the mathematical simulations. The stress response function of Feddes et al. (1978) and the S-shaped logistic root growth function were utilized. The values for P_0, $P_{2\ L}$, $P_{2\ H}$, P_3 and P_{opt} were set to −10, −200, −800, −8000 and −25 cm, respectively. The initial root growth time was set at the 115th Julian day and the harvest time on the 242th Julian day. The minimum and maximum rooting depths were set to 0.02 and 90 cm, respectively.

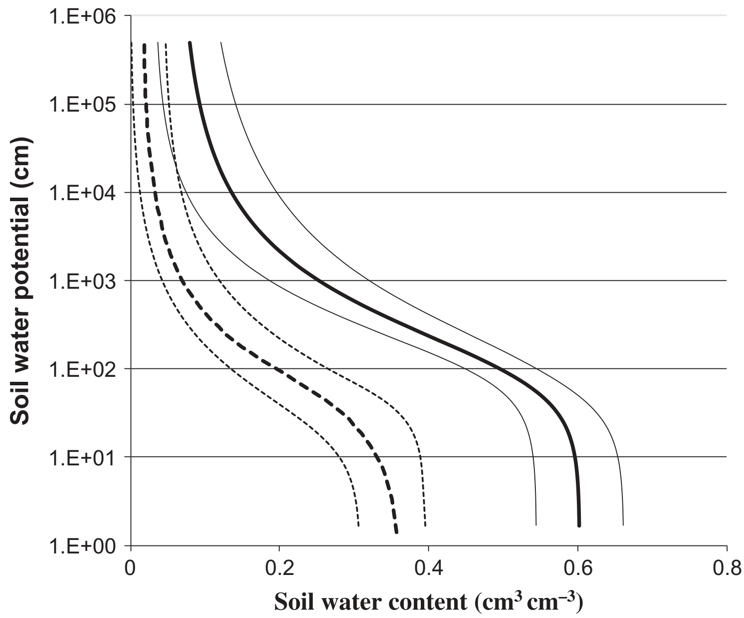

Figure 2. Statistically processed soil water retention curves for two textural classes. The three dashed curves on the left relate to loamy sand and the three solid curves relate to clay loam. The bold lines are the average curves, and the thin lines are the contour curves created by adding or subtracting the value of standard deviation from the average value.

Results

Collected SWC field data

The time series of precipitation and groundwater level (boundary water inputs) in the study field during the investigation period 2002–2003 are shown in Figures 3 and 4.

Despite 16 days without rain preceded soil sampling, the soil profile was sufficiently well supplied by groundwater discharge (capillary rise) on 10 April 2002. The groundwater level was only 0.7 m below the surface. The average saturation of soil was 64% (87% for clay loam and 46% for loamy sand). The values of actual SWC showed a bimodal distribution (Figure 5): the mean value in clay loam was 45% (SD 5%), and in loamy sand was 16% (SD 4.5%).

SWC was again measured across the experimental field on 2 July 2003 to examine the soil moisture conditions under high evaporative atmospheric demand (5 mm day^{-1}). Total precipitation was 102 mm over the three preceding months. After the spring snow had thawed, the groundwater table had dropped to a level of 2.39 m below the soil surface in July 2003 (Figure 4). The mean value for clay loam was 37% (SD 8%) and for loamy sand was 11% (SD 3.9%). SWC data relating to soil texture had normal distribution confirmed by the Kolmogorov–Smirnov test, at a *p*-level of 0.1.

Spatial distribution of SWC on these two sampling dates with respect to the critical θ_{la} calculated for ET rates of 1 and 5 mm, according to Equations (1)–(3) (Table 1), was analyzed in Orfanus and Eitzinger (2010). In this paper we focused on the temporal changes of SWC variability during the spring–summer period in 2003, which was characterized by typical lowering of the groundwater table with the riverine regime (Figure 4).

Results of HYDRUS–ET performance testing

To evaluate the temporal changes in variability of SWC with respect to the main drying–wetting processes during the period 16 February–2 July 2003, we decided to use a mathematical model

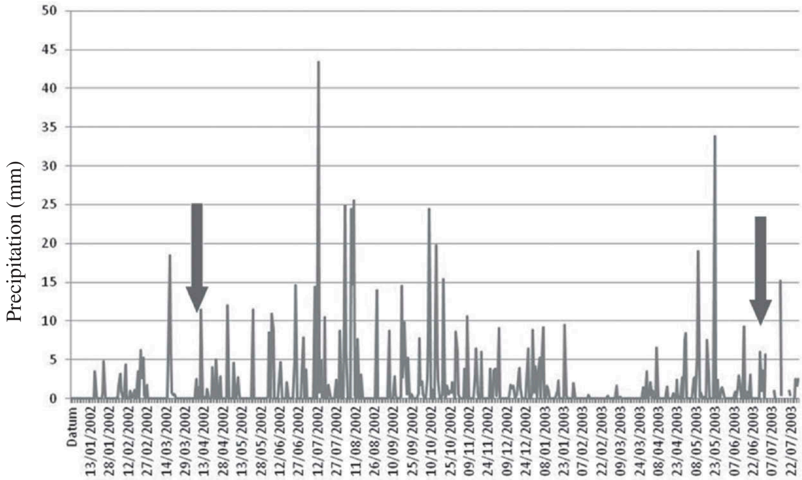

Figure 3. Precipitation data from the climatological station in Moravsky Sv. Jan. The soil moisture sampling dates on 10 April 2002 and 2 July 2003 are demarcated by arrows.

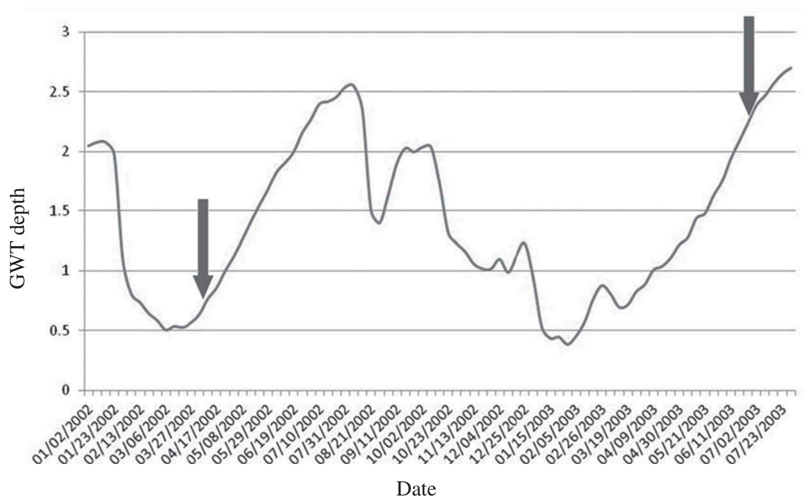

Figure 4. Groundwater table position (in meters below ground surface) during the period 2002–2003. The soil moisture sampling dates on 10 April 2002 and 2 July 2003 are demarcated by arrows.

rather than laborious recurrent SWC sampling. Mathematical simulations with the HYDRUS–ET model were performed following the algorithm described in Materials and methods. The input hydrophysical characteristics for particular simulations together with the results (for 10 April 2002) are shown in Tables 2 and 3.

The mathematical model performed very well on our four soil profiles, with precisely determined topsoil and subsoil hydrophysical characteristics. Errors were less than 4% vol. (Table 4).

For measurements carried out on 10 April 2002 and 2 July 2003 the simulations overestimated SWC distribution by 3–6% in regard to mean values when compared to gravimetrically measured data from the soil horizon at 10–15 cm depth.

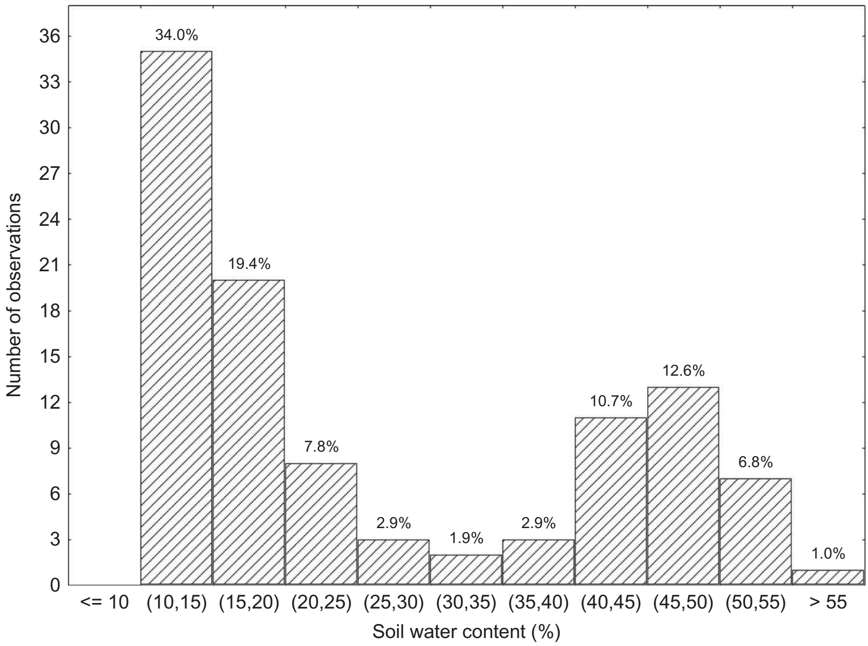

Figure 5. Bimodal distribution of SWC data over the 4.5 ha field follows its textural heterogeneity.

Table 1. Descriptive statistics of SWC of limited availability estimated for two textural classes and two evapotranspiration rates, in MSJ field experimental site.

	Mollic Gleysol		Arenic Regosol	
Soil type	Clay loam		Loamy sand	
Texture	θ_{la} ($E_{p\text{-max}}$)	θ_{la} ($E_{p\text{-min}}$)	θ_{la} ($E_{p\text{-max}}$)	θ_{la} ($E_{p\text{-min}}$)
Mean	0.249	0.152	0.161	0.089
Standard error	0.009	0.010	0.006	0.006
Median	0.230	0.133	0.149	0.010
Standard deviation	0.042	0.042	0.026	0.026
Sample variance	0.002	0.002	0.001	0.001
Kurtosis	0.568	0.568	0.069	0.769
Skewness	1.156	1.156	1.368	1.368
Confidence level (95%)	0.020	0.020	0.012	0.012

Table 2. Numerical simulation arrangement for loamy sand part of the MSJ field.

SWRC		Volumetric soil water content														
Depth		K (cm day^{-1})														
10–15 cm	30–35 cm	12	14	19	25	29	34,5	39	46	50	57	70	100	150	200	250
	Mean	16.4	16.6	16.8	17.0	17.2	17.5	17.7	18.0	18.3	18.3	19.2	19.2	19.3	19.6	19.9
Mean	Mean + SD	16.8	16.8	17.0	17.4	17.7	18.3	18.6	19.0	19.0	19.2	19.4	19.5	19.5	19.9	20.3
	Mean − SD	15.8	15.8	16.1	16.2	16.4	16.4	16.7	17.0	17.7	18.1	19.0	19.2	19.2	19.3	19.6
	Mean	20.0	20.6	21.6	22.1	23.0	23.3	23.5	23.7	23.8	23.8	24.1	24.1	24.2	24.4	24.6
Mean+SD	Mean + SD	20.8	20.8	21.5	22.8	23.5	23.6	23.7	23.8	23.8	24.0	24.5	24.6	24.6	24.8	24.8
	Mean − SD	18.9	18.9	20.0	20.7	21.3	21.3	21.6	22.7	23.0	23.2	24.0	24.0	24.0	23.8	23.8
	Mean	15.0	15.0	15.0	15.0	15.0	15.3	15.5	15.8	16.2	16.2	17.3	17.3	17.4	17.7	18.1
Mean − SD	Mean + SD	15.2	15.2	15.2	15.3	15.4	16.2	16.6	17.1	17.2	17.3	17.5	17.5	17.5	18.0	18.5
	Mean − SD	14.6	14.6	14.6	14.5	14.5	14.5	14.8	14.8	15.7	16.1	17.1	17.3	17.4	17.6	18.0

Different combinations of hydrophysical characteristics: SWRC – soil water retention curve; K – saturated hydraulic conductivity, are provided together with simulated SWC values for 10 April 2002.

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Table 3. Numerical simulation arrangement for clay loam part of the MSJ field.

SWRC		Volumetric soil water content														
Depth		K (cm day^{-1})														
10–15 cm	30–35 cm	0.50	1	2	3	5	7	9	16	22	26	30	42	68	77	106
	Mean	42.6	43.6	44.6	46.0	46.0	47.2	48.3	49.3	51.4	51.4	51.3	51.3	51.3	51.3	51.3
Mean	Mean + SD	42.2	43.2	44.2	45.0	45.9	47.8	48.8	49.6	51.3	51.3	51.3	51.2	51.2	51.2	51.2
	Mean − SD	43.2	44.1	45.0	46.1	46.8	47.4	48.5	49.6	51.4	51.4	51.4	51.3	51.3	51.3	51.3
	Mean	47.5	48.0	48.4	49.1	50.3	51.1	52.3	53.5	54.8	54.8	54.7	54.7	54.7	54.7	54.7
Mean+SD	Mean + SD	47.1	47.7	48.1	48.7	50.0	51.7	52.9	53.4	54.7	54.7	54.7	54.7	54.7	54.7	54.7
	Mean − SD	47.4	48.1	48.7	49.2	50.3	51.4	52.5	53.6	54.8	54.8	54.7	54.7	54.7	54.7	54.7
	Mean	40.2	41.5	42.8	43.9	44.5	45.3	46.4	47.3	49.8	49.8	49.7	49.7	49.7	49.7	49.7
Mean −SD	Mean + SD	39.8	41.0	42.3	43.3	44.0	45.9	46.8	47.8	49.8	49.7	49.7	49.5	49.5	49.6	49.6
	Mean − SD	41.2	42.2	43.3	44.6	45.1	45.5	46.6	47.7	49.8	49.8	49.8	49.8	49.7	49.7	49.7

Different combinations of hydrophysical characteristics: SWRC – soil water retention curve; K – saturated hydraulic conductivity, are provided together with simulated SWC values for 10 April 2002.

Table 4. Verification of the HYDRUS–ET model on four soil profiles for 10 April 2002.

Test soil	Textural class	Measured SWC (%)	Simulated SWC (%)
Mollic Gleysol	Clay loam 1	49.8	47.7/4.2
	Clay loam 2	39.2	43.1/10.1
Arenic Regosol	Loamy sand 1	12.5	12.2/2.6
	Loamy sand 2	10.8	9.6/10.7

We suppose that this shift in mean SWC values was the result of ignoring hysteresis in the soil water retention curve in the model. Swelling–shrinking of clay loam could also have contributed to higher (5%) overestimation of SWC in this part of the field. However, otherwise the SWC distributions were simulated well. The normal distributions of simulated data were not rejected by the Kolmogorov–Smirnov test at p-level = 0.1; SD in loamy sand and clay loam, respectively was 3.2 and 3.8% on 10 April 2002 and 3.7 and 6.8% on 2 July 2003.

SWC variability during the simulation period

Substantial increase in SWC variability from the beginning of June to 2 July 2003 (Figure 6) in the clay loam part of the field can be ascribed to drainage-induced variability, which can be explained as follows. When soil is wet there is no limit to potential evaporation and the evaporation rate is close to an atmospherically controlled potential rate that usually shows relatively small spatial variation (Pan & Peters-Lidard 2008). Under such conditions the variability of SWC is controlled by drainage as the groundwater table progressively falls.

However, after reaching the threshold SWC (θ_{la}) the evaporation rate is no longer controlled by atmospheric conditions only, and the actual evaporation rate becomes proportional to SWC. At this stage SWC variability decreases as mean SWC decreases (due to selective evapotranspiration). This was the case for loamy sand (Figure 6) during the period between 4 June and 2 July 2003, when in most locations SWC decreased under the critical (θ_{la}). Here the mean SWC had decreased and the variability (SD) also decreased. Our observations – as well as those of Zhao et al. (2010) – are in agreement with the theory that in relatively dry soil, drainage is a negligible drying factor when compared to evapotranspiration, the wetter parts evaporating more water than the drier parts and variability lowered with decreasing SWC mean value. This relationship was also well demonstrated by Vivoni et al. (2008) for different semiarid ecosystems, and theoretically proven by Pan & Peters-Lidard (2008).

Results from 2 July 2003 show that the SWC in loamy sand was lower than the θ_{la} estimated for high evapotranspiration rates over practically the whole area (Figure 7). On the other hand, in clay

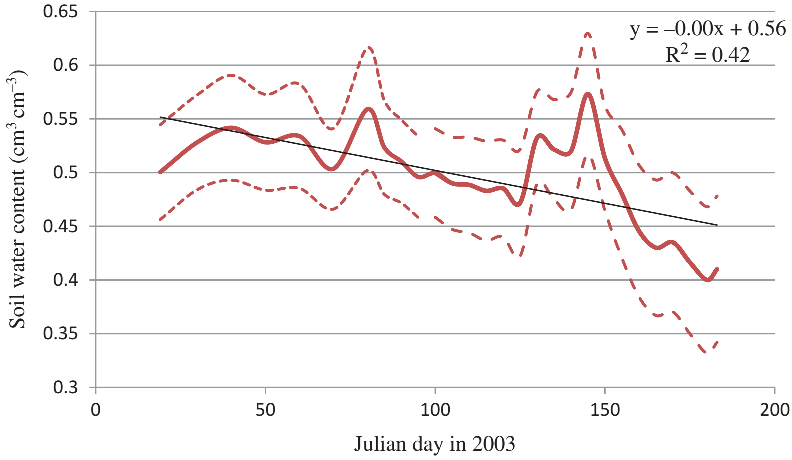

Figure 6. The course of SWC and its variability in the clay loam part of the MSJ field. The curves denote the mean (bold, solid) and mean +/− SD (dashed) soil water contents on particular days of the spring–summer period in 2003. The thin line is the linear descending trend.

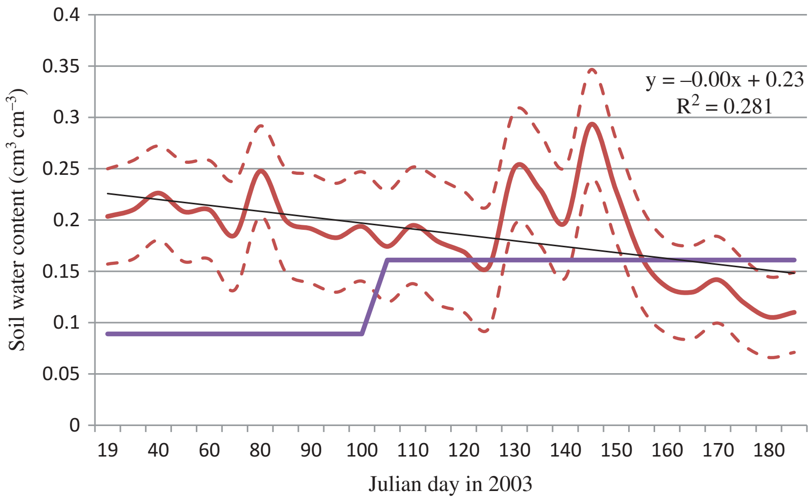

Figure 7. The course of SWC and its variability in loamy sand part of the MSJ field. The curves denote the mean (bold solid) and mean +/− SD (dashed) SWC on particular days of the spring–summer period in 2003. The thin line is the linear descending trend and the bold line is the critical SWC of limited availability, θ_{la}. θ_{la} is configured for the minimum value of ETP = 1 mm (from the beginning of the year to the 105th Julian day) and maximum value of ETP = 5 mm (from the 105th to the 183rd Julian day).

loam it was lower only sporadically. The mean θ_{la} here was 0.25, which was not matched by any SWC value from the range mean ± SD (Figure 6).

The shallow spring water table in 2003 had decreased steadily to a depth of 2.39 m (Figure 3) by 2 July 2003. Capillary rise from groundwater was prevented in loamy sand, and a substantial volume of soil water from the horizon at a depth of 0.10–0.15 m had drained into deeper layers. In clay loam soil the available water reserves were maintained much longer than in loamy sand because the water storage capacity of texturally finer soil is much higher. Moreover, the water depleted by ET was still to some extent compensated by capillary rise from the groundwater in the clay loam part of the field, while water losses through downward infiltration are very rapid in coarse-textured soil (Starr 2005; Pan & Peters-Lidard 2008).

Conclusions

The influence of the natural heterogeneity of soil texture on SWC variability was studied in the village of Moravsky Sv. Jan (MSJ), Zahorska Lowland, western Slovakia. The concept of temporally variable critical SWC (θ_{la}), below which the drainage of soil ceases and relative evapotranspiration drops below 1, was used as an indicator when the relationship between mean SWC and its standard deviation reverses. SWC lower than θ_{la} indicates that the only drying process active at this stage is evapotranspiration, which begins to be selective, i.e. the wetter parts of the field evaporate more than the drier parts.

The SWC measurement campaigns in the MSJ experimental field, on 10 April 2002 and 2 July 2003, together with mathematical simulations of SWC dynamics during the period 16 February–2 July 2003, which cover variable climate and groundwater situations, revealed that:

(1) At the stage when evapotranspiration is not limited by SWC (i.e. SWC $\gg \theta_{la}$) and drainage of the soil profile is prevented by a shallow groundwater table, the variability of SWC in the soil horizon 0.10–0.15 m was low. This was the case for the first sampling date (10 April 2002) when the variability of SWC in both soil textures was controlled by capillary recharge from shallow groundwater (0.77 m depth), saturating the soil profile far above the value of field capacity.
(2) Under the condition where drainage was active (SWC > θ_{la}, deep groundwater) and prevailed over evapotranspiration (negative flux in HYDRUS–ET simulations), the texturally coarser parts of the field drained faster than the finer parts and SWC variability increased with decreasing mean SWC over the field. This was demonstrated in loamy sand between the 1st and the 135th days and in the clay loam part between the 100th and 183rd days of the mathematical simulation in 2003.
(3) As drying of soil continues below θ_{la}, the drainage of soil profile ceases while the ET becomes the major drying process and the variability of SWC undergoes the opposite (decreasing) trend with decreasing mean SWC. This is because ET is limited and the wetter parts of the field evaporate more water than drier parts, smoothing SWC variability. We observed this in loamy sand from results of mathematical simulation between the 135th and 183rd days of 2003.

The principles of SWC variability identified in this study can be applied to irrigation management. The highest spatial variability of SWC at the field scale can be expected at dates and at locations when (where) there has been significant drainage of the soil profile but where there is still enough water to supply the rate of ET equal to the potential rate. At these stages and at target locations the point-source precise irrigation dosages can be applied with highest efficiency while the ongoing drying of the field under the critical SWC will smooth SWC variability, information about water deficiency over the field will be lost and abundant wide-scale irrigation would have to be applied.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

Bell KR, Blanchard BJ, Schmugge TJ, Witczak MW. 1980. Analysis of surface moisture variations within large-field sites. Water Resour Res. 16:796–810.

Cowan IR. 1965. Transport of water in the soil-plant-atmosphere system. J Appl Ecol. 2:221–239.

Feddes RA, Kowalik PJ, Zaradny H. 1978. Simulation of filed water use and crop yield. Simulation monographs. Wageningen: Pudoc.

Francis CF, Thornes JB, Romero Diaz A, Lopez Bermudez F, Fisher GC. 1986. Topographic control of soil moisture, vegetation cover and land degradation in a moisture stressed mediterranean environment. Catena. 13:211–225.

Georgakakos K, Baumer OW. 1996. Measurement and utilization of on-site soil moisture data. J Hydrol. 184:131–152.
Grayson RB, Western AW, Chiew FHS. 1997. Preferred states in spatial moisture soil patterns: local and nonlocal controls. Water Resour Res. 33:2897–2908.
Hawley ME, Jackson TJ, McCuen RH. 1983. Surface soil moisture variation on small agricultural watersheds. J Hydrol. 62:179–200.
Hillel D. 1998. Environmental soil physics. San Diego, CA: Academic Press.
Hills RC, Reynolds SG. 1969. Illustrations of soil water content variability in selected areas and plots of different sizes. J Hydrol. 8:27–47.
Idso SB, Reginato RJ, Jackson RD, Kimball BA, Nakayama FS. 1974. The three stages of drying of a field soil. Soil Sci Soc Am Proc. 38:831–837.
ISSS-ISRIC-FAO. 1998. World reference base for soil resources. World Soil Resources Reports 84. Rome: Food and Agriculture Organization of the United Nations.
Loague K. 1992. Soil water content at R-5. Part 1. Spatial and temporal variability. J Hydrol. 139:233–251.
Nielsen DR, Biggar JW, Erh KT. 1973. Spatial variability of field-measured soil-water properties. Hilgardia. 42:215–259.
Novák V. 1986. Stanovenie odberu vody koreňmi rastlín z poľných meraní [Plant root water uptake determination from field measurements]. Vodohosp Čas. Bratislava: Institute of Hydrology, Slovak Academy of Sciences; p. 314–329.
Novák V. 1989. Výpočet denných úhrnov evapotranspirácie použitím modifikovanej Penmannovej metódy [Calculation of daily evapotranspiration sums using modified Penmann method]. Vodohosp Čas. Bratislava: Institute of Hydrology, Slovak Academy of Sciences; p. 113–129.
Nyberg L. 1996. Spatial variability of water content in covered catchment at Gardsjon, Sweden. Hydrol Process. 10:89–103.
Orfanus T, Eitzinger J. 2010. Factors influencing the occurrence of water stress at field scale. Ecohydrology. 3:478–486.
Pan F, Peters-Lidard CD. 2008. On the relationship between mean and variance of soil moisture fields. J Am Water Resour As (JAWRA). 44:235–242.
Šimůnek J, Huang K, Sejna M, Genuchten Van M, Novak V, Sutor J. 1997. The HYDRUS-ET software package for simulating the one-dimensional movement of water, heat, and multiple solutes in variably-saturated media. Version 1.1. Bratislava: Slovak Academy of Sciences.
Starr GC. 2005. Assessing temporal stability and spatial variability of soil water patterns with implications for precision water management. Agric Water Manage. 72:223–243.
Sudnicyn II. 1979. Water flow in soils and use of water by plants. Russian: Moscow. Izd. MGU .
van Genuchten M. 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. Soil Sci Soc Am J. 44:892–898.
van Genuchten MTh. 1985. Convective-dispersive transport of solutes involved in sequential first-order decay reactions. Computers & Geosci. 11:129–147.
Vivoni ER, Rinehart AJ, Méndez-Barroso LA, Aragón CA, Bisht G, Cardenas MB, Engle E, Forman BA, Frisbee MD, Gutiérrez-Jurado HA, et al. 2008. Vegetation controls on soil moisture distribution in the Valles Caldera, New Mexico, during the North American monsoon. Ecohydrology. 1:225–238.
Warrick AW, Mullen GJ, Nielsen DR. 1977. Predictions of the soil water flux based upon field-measured soil-water properties. Soil Sci Soc Am J. 41:14–19.
Wesseling JG, Brandyk T. 1985. Introduction of the occurrence of high groundwater levels and surface water storage in computer program SWATRE. Wageningen, The Netherlands: Institute for Land Water Management Research.
Western AW, Blöschl G. 1999. On the spatial scaling of soil water content. J Hydrol. 217:203–224.
Western AW, Blöschl G, Grayson RB. 1998. Geostatistical characterisation of soil moisture patterns in the Tarrawarra catchment. J Hydrol. 205:20–37.
Western AW, Grayson RB. 1998. The Tarrawarra data set: soil moisture patterns, soil characteristics, and hydrological flux measurements. Water Resour Res. 34:2765–2768.
Zhao WZ, Liu B, Zhang ZH. 2010. Water requirements of maize in the middle Heihe River basin, China. Agric Water Manage. 97:15–223.